



How Colour Laser printers work

The technology behind a laser printer is completely different from that of inkjets. A laser printer has a laser beam that creates an image of what you're printing, on the photoreceptive drum. Here's how a laser printer works:



Shown here are the four toner cartridges that are responsible for the colour, and the toner drum with Photoreceptor roller (green roller).

The PC sends printed matter in either HP's PCL (Printer Command Language) or Adobe's PostScript language. Here the data is converted into vector information.

The printer converts the vector information into dots (or, a bitmap). Some printers use the GDI (Graphical Digital Interface), where the bitmap is generated by the PC, and then passed on to the laser.

Now, the laser fires when there should be a dot, and remains



This figure shows the holder, where the toner cartridge are seated. The gear, which is motor driven, helps in rotating the toner cartridge.

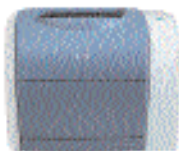
off when there should be a blank space. The onboard electronics control the movement of the laser, and the beam is then passed through the lens filter to the Photoreceptor drum. Here the laser beam is moved horizontally only, hence creating the dot patterns on the photoreceptor drum. Once an entire line of dots is completed, the drum rotates to receive the next line. Wherever the laser hits the positively-charged drum, a negative charge is created. The toner consists of a pigment that imparts colour onto a paper. The pigment is blended into a plastic particle to prevent smudging and making the borders of images look sharp. There are four toners in a colour laser printer—cyan, magenta, yellow and black. The positively charged toner attaches itself to the negatively charged regions of the drum.



Shown above is the drum cartridge that holds the photoreceptor drum and corona wires. On the opposite side is the Cyan toner cartridge, which also holds the toner developer and toner hopper.

This is done four times in a colour laser printer. Now the paper is passed over the drums. The paper is negatively charged by the transfer corona wires. This charge is greater than the negative charge on the drum, and thus attracts the positively charged toner pigment. The pigment doesn't stick to the paper here, and the paper is then neutralised and passed through the fuser assembly—nothing but a Teflon coated metal roller, heated by the quartz lamp. The heat fuses the toner and the paper—this is why if you print multiple copies one after another, the output paper becomes increasingly warmer. The pattern is cleaned off the drum using the charged corona wire, and the printer is ready for the next document.

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